CSCA67/MATA67 Week 1

CSCA67/MATA67 Week 1

Discrete Mathematics

Anna Bretscher

Combinatorics aka Counting

Combinatorics aka Counting

Helps us answer questions like…

Combinatorics aka Counting

Helps us answer questions like…

Combinatorics aka Counting

Helps us answer questions like…

Q. How much more secure is a 6 digit versus 4 digit phone phone passcode?

Combinatorics aka Counting

Combinatorics aka Counting

Helps us answer questions like…

Combinatorics aka Counting

Helps us answer questions like…

Combinatorics aka Counting

Helps us answer questions like…

Q. Given 10 flavours of donuts at Tim Hortons, how many ways can you select a dozen donuts?

Combinatorics aka Counting

Combinatorics aka Counting

Helps us answer questions like…

Combinatorics aka Counting

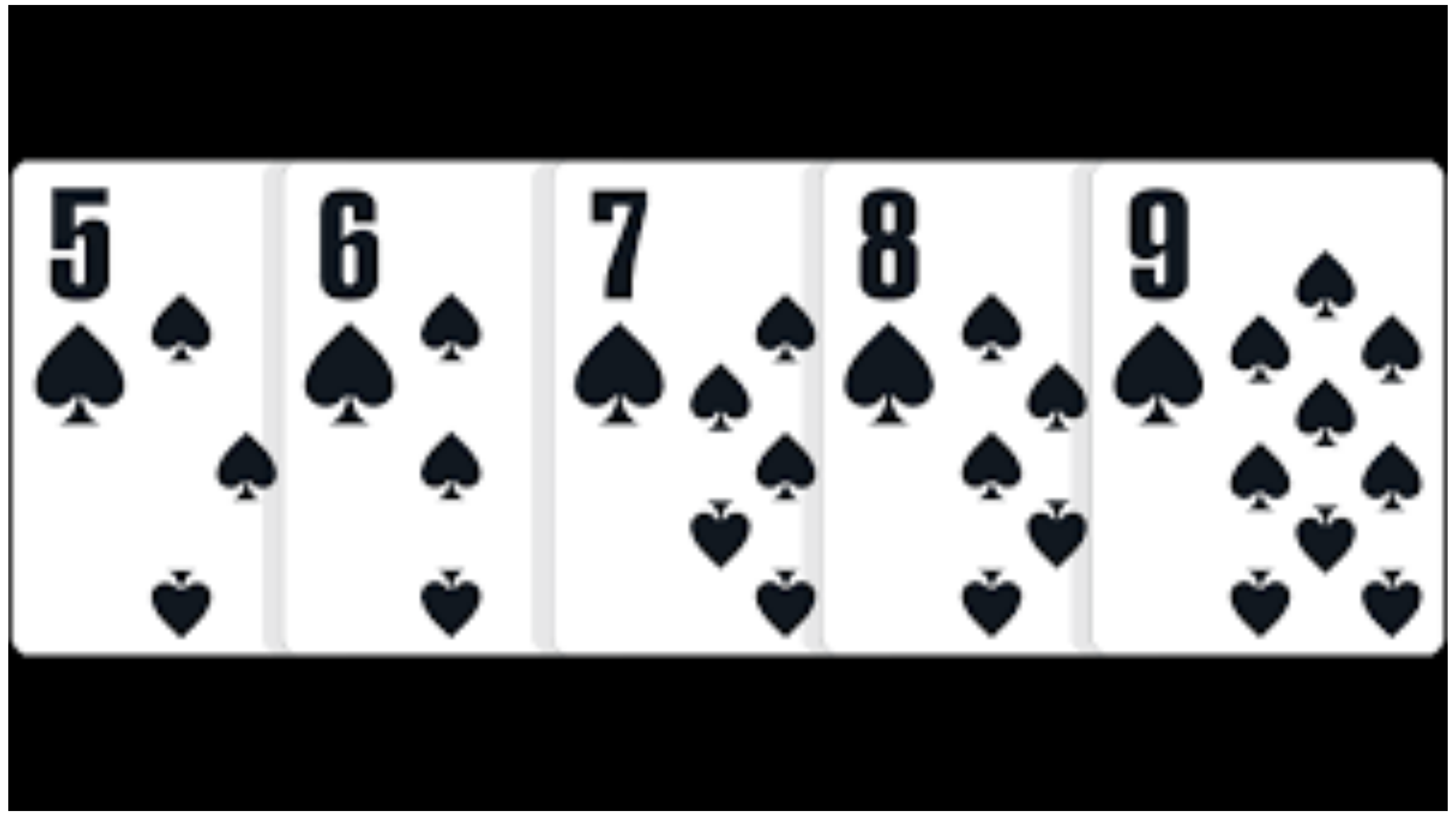
Helps us answer questions like…

Combinatorics aka Counting

Helps us answer questions like…

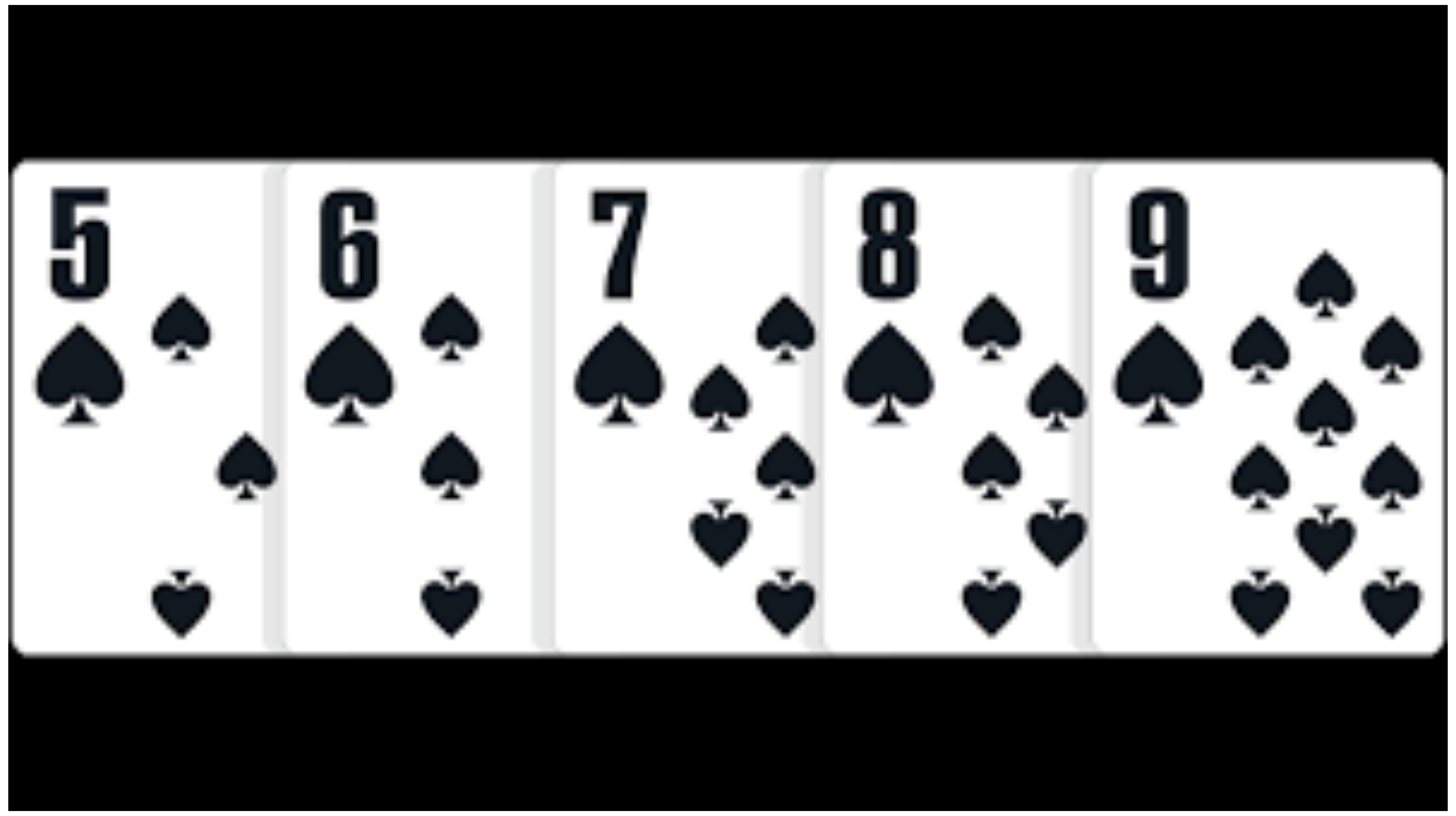
Q. In the 2020 Olympic 100m freestyle swim finals, how many different orders can the 8 competitors finish?

Combinatorics aka Counting



Combinatorics aka Counting

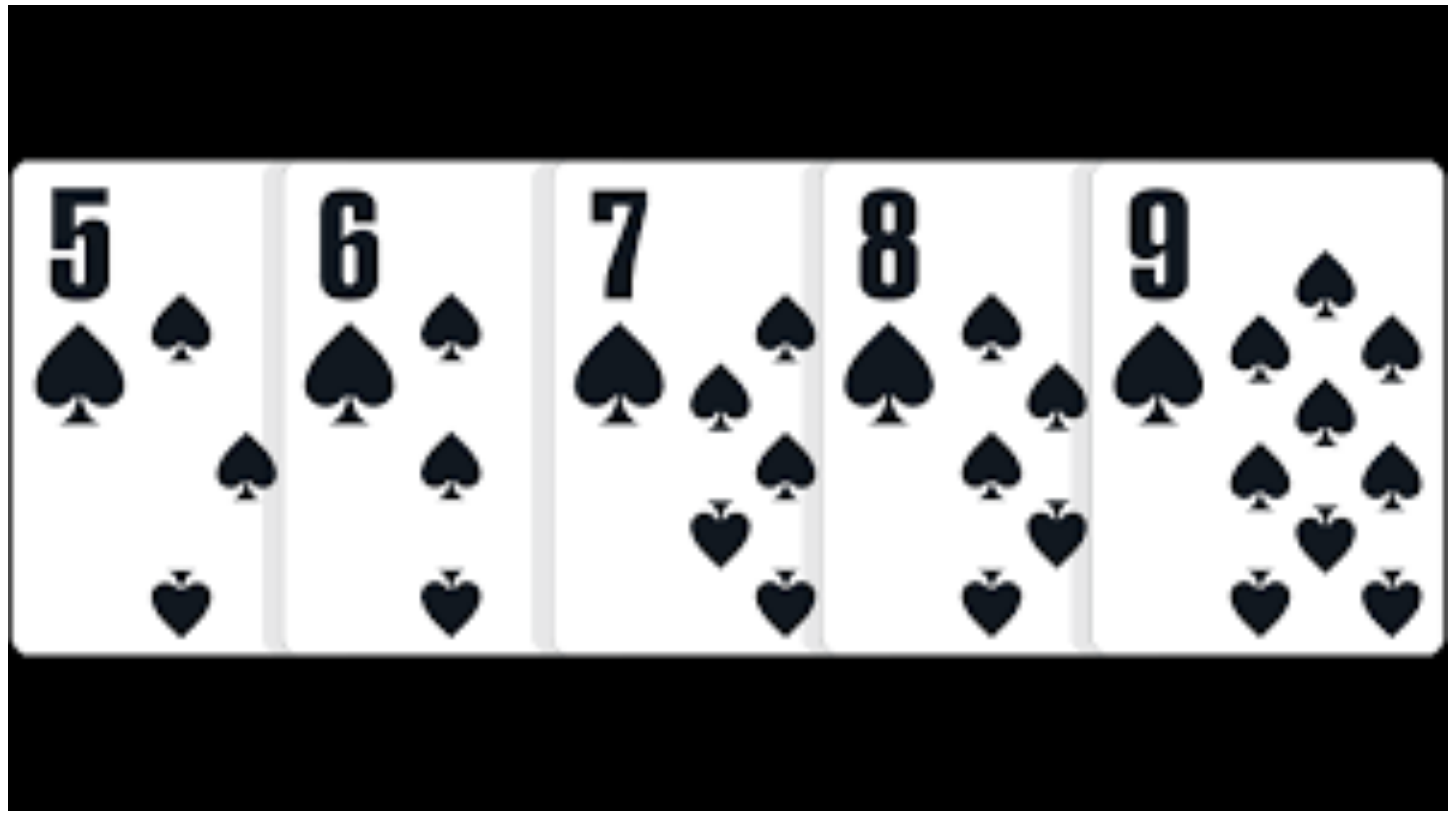
Helps us answer questions like…



Combinatorics aka Counting

Helps us answer questions like…

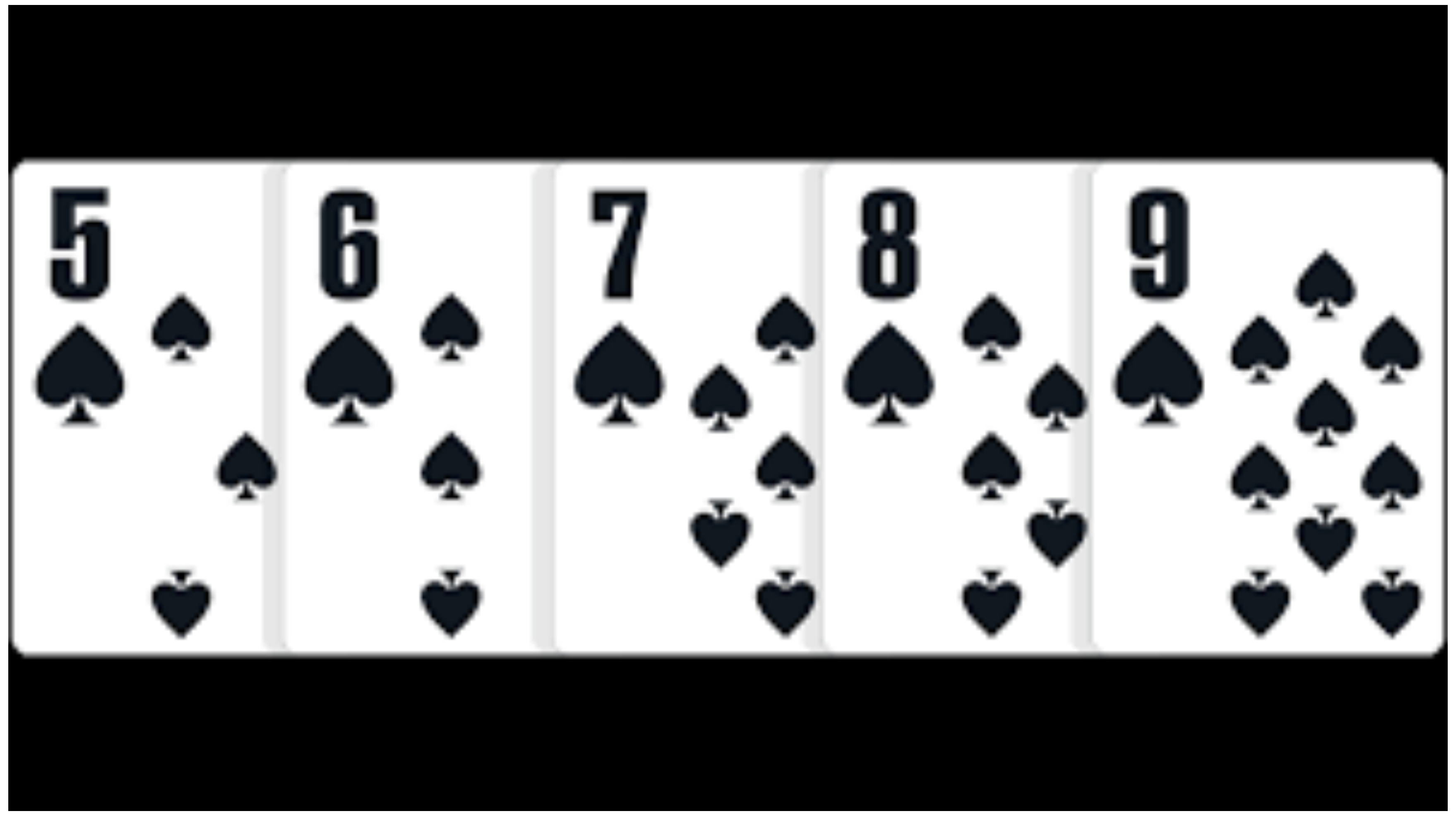
Q. In poker, how many different 5 card hands can you have?



Combinatorics aka Counting

Helps us answer questions like…

Q. In poker, how many different 5 card hands can you have?



Q. How many different ways are there to get a straight flush?

Order and Repetition

Order and Repetition

We can try to solve these problems by first asking *two questions.*

Order and Repetition

We can try to solve these problems by first asking *two questions.*

Q1. Does order matter?

Order and Repetition

We can try to solve these problems by first asking *two questions.*

Q1. Does order matter?

Q2. Is there repetition?

For which problems did order matter?

Order and Repetition

We can try to solve these problems by first asking *two questions.*

Q1. Does order matter?

Q2. Is there repetition?

For which problems did order matter?

For which problems was there repetition?

Permutations vs Combinations

Permutations vs Combinations

We can try to solve these problems by first asking *two questions.*

Permutations vs Combinations

We can try to solve these problems by first asking *two questions.*

If order matters then:

➡ *We are counting the number of permutations.*

Permutations vs Combinations

We can try to solve these problems by first asking *two questions.*

If order matters then:

➡ *We are counting the number of permutations.*

If order does not matter then:

➡ *We are counting the number of selections or combinations.*

Permutations vs Combinations

We can try to solve these problems by first asking *two questions.*

If order matters then:

➡ *We are counting the number of permutations.*

If order does not matter then:

➡ *We are counting the number of selections or combinations.*

For which problems did order matter?

Permutations vs Combinations

We can try to solve these problems by first asking *two questions.*

If order matters then:

➡ *We are counting the number of permutations.*

If order does not matter then:

➡ *We are counting the number of selections or combinations.*

For which problems did order matter?

For which problems was there repetition?

Permutations vs Combinations

	Order Matters	Order Does Not Matter
Repetition Allowed	Phone passcode	Donuts
No Repetition	Swim Race Outcome	Poker straight flush

Permutation

Combination

Permutations vs Combinations

Permutations vs Combinations

Let's practice!

Passcodes

Passcodes

Q. How many 4 digit phone passcodes are possible?

Passcodes

Q. How many 4 digit phone passcodes are possible?

A. 10^4

Q. How many 6 digit phone passcodes are possible?

Passcodes

Q. How many 4 digit phone passcodes are possible?

A. 10^4

Q. How many 6 digit phone passcodes are possible?

A. 10^6

Permutation with Repetition

Count the number of ways to arrange *4 objects* or *6 objects* from *10 choices* with *repetition allowed* and *order matters.*

Permutation with Repetition

Count the number of ways to arrange *4 objects* or *6 objects* from *10 choices* with *repetition allowed* and *order matters.*

$\Rightarrow$ gives 10^4 or 10^6 possible permutations.

Permutation with Repetition

Count the number of ways to arrange *4 objects* or *6 objects* from *10 choices* with *repetition allowed* and *order matters.*

$\Rightarrow$ gives 10^4 or 10^6 possible permutations.

Take Home

Permutation with Repetition

Count the number of ways to arrange *4 objects* or *6 objects* from *10 choices* with *repetition allowed* and *order matters.*

$\Rightarrow$ gives 10^4 or 10^6 possible permutations.

Take Home

The formula to count *r* objects *from n choices with repetition* when *order matters* is: n^r

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

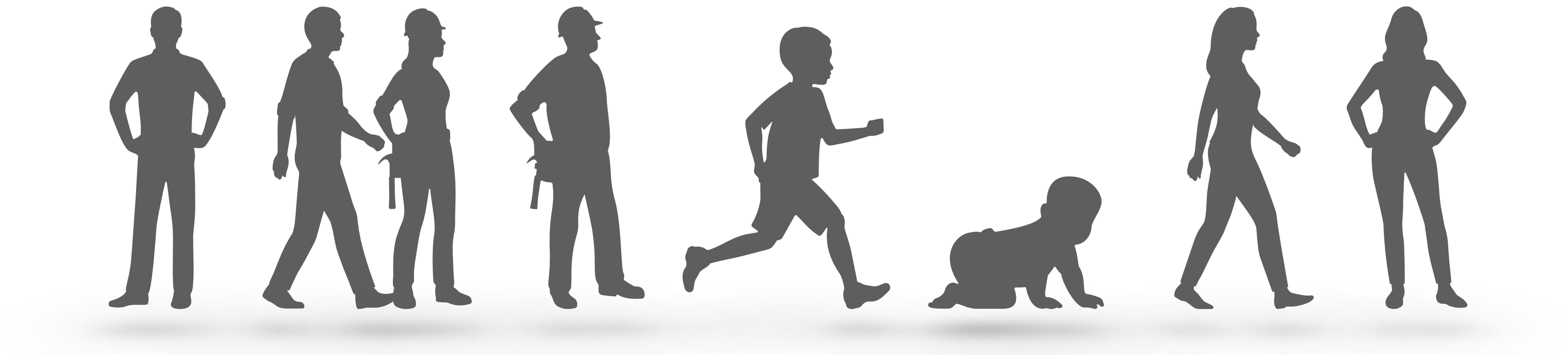
Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

Q. *Does order matter? Is there repetition?*

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?



Q. *Does order matter? Is there repetition?*

A. Yes! And no!

How do we solve this?

__ __ __ __ __ __ __ __

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

Q. *Does order matter? Is there repetition?*

A. Yes! And no!

How do we solve this? __ __ __ __ __ __ __ __

How many choices for the first person? 8 __ __ __ __ __ __ __

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

Q. Does order matter? Is there repetition?

A. Yes! And no!

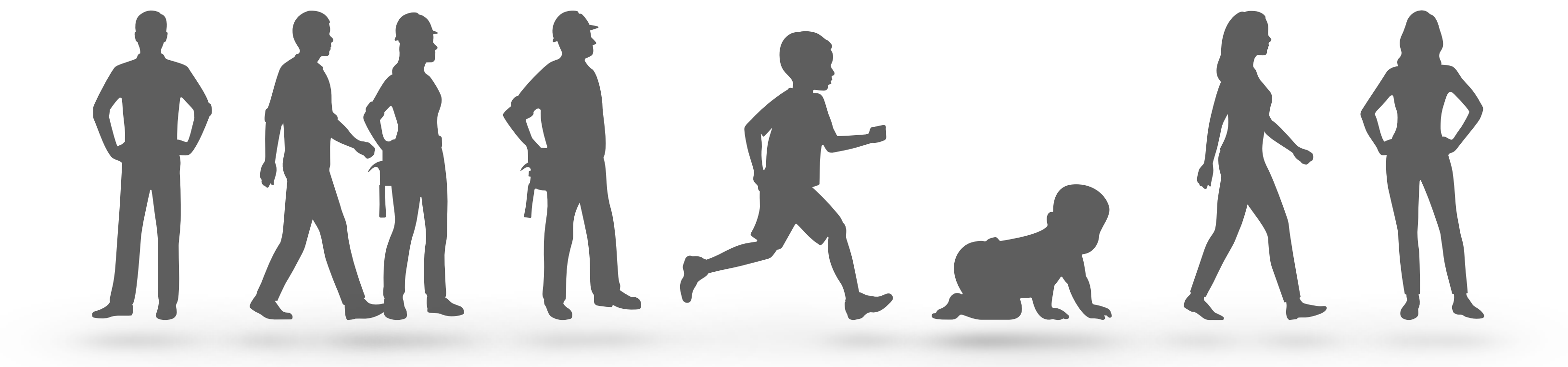
How do we solve this? __ __ __ __ __ __ __ __

How many choices for the first person? 8 __ __ __ __ __ __ __

How many choices for the second person? 8 * 7 __ __ __ __ __ __

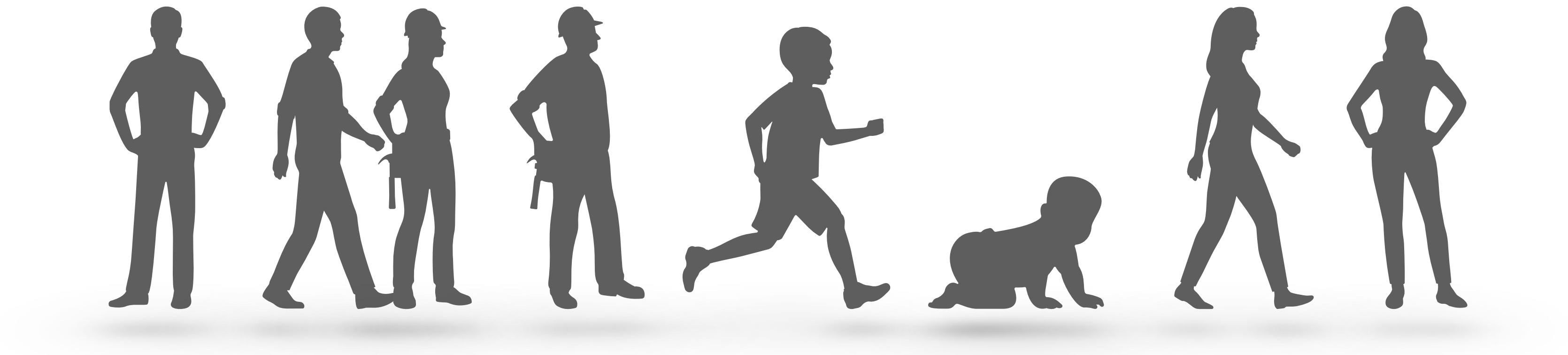
Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?



Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?



If we continue this pattern, we get:

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

If we continue this pattern, we get:

$$8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 8!$$

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

If we continue this pattern, we get:

8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 8!

Take Home

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

If we continue this pattern, we get:

8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 8!

Take Home

The formula to arrange *n* objects *without repetition* when *order matters* is: n!

Service at the bank

8 people lineup at a bank. In how many *different orders* can they be served?

Why do we multiply instead of add?

If we continue this pattern, we get:

$$8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 8!$$

Take Home

The formula to arrange *n* objects *without repetition* when *order matters* is: n!

Product Rule

Product Rule

Product Rule

Product Rule

Product Rule

Suppose a procedure can be broken down into a sequence of k tasks.

Product Rule

Product Rule

Suppose a procedure can be broken down into a sequence of k tasks.

Product Rule

Product Rule

Suppose a procedure can be broken down into a sequence of k tasks.

If the i^{th} task takes n_i steps then there are $n_1 n_2 n_3 \cdots n_k$ ways to do the procedure.

Product Rule

Product Rule

Suppose a procedure can be broken down into a sequence of k tasks.

If the i^{th} task takes n_i steps then there are $n_1 n_2 n_3 \cdots n_k$ ways to do the procedure.

Product Rule

Why do we multiply instead of add?

Product Rule

Suppose a procedure can be broken down into a sequence of k tasks.

If the i^{th} task takes n_i steps then there are $n_1 n_2 n_3 \cdots n_k$ ways to do the procedure.

Sum Rule

Sum Rule

Sum Rule

Sum Rule

Sum Rule

If a task can be done in one of k different ways then the sum rule applies.

Sum Rule

Sum Rule

If a task can be done in one of k different ways then the sum rule applies.

Sum Rule

Sum Rule

If a task can be done in one of k different ways then the sum rule applies.

If the i^{th} way takes n_i steps then there are $n_1 + n_2 + n_3 + \cdots + n_k$ ways to do the task.

Sum Rule

Sum Rule

If a task can be done in one of k different ways then the sum rule applies.

If the i^{th} way takes n_i steps then there are $n_1 + n_2 + n_3 + \cdots + n_k$ ways to do the task.

Q. Can you think of an example that uses the sum rule?

Sum Rule

Sum Rule

If a task can be done in one of k different ways then the sum rule applies.

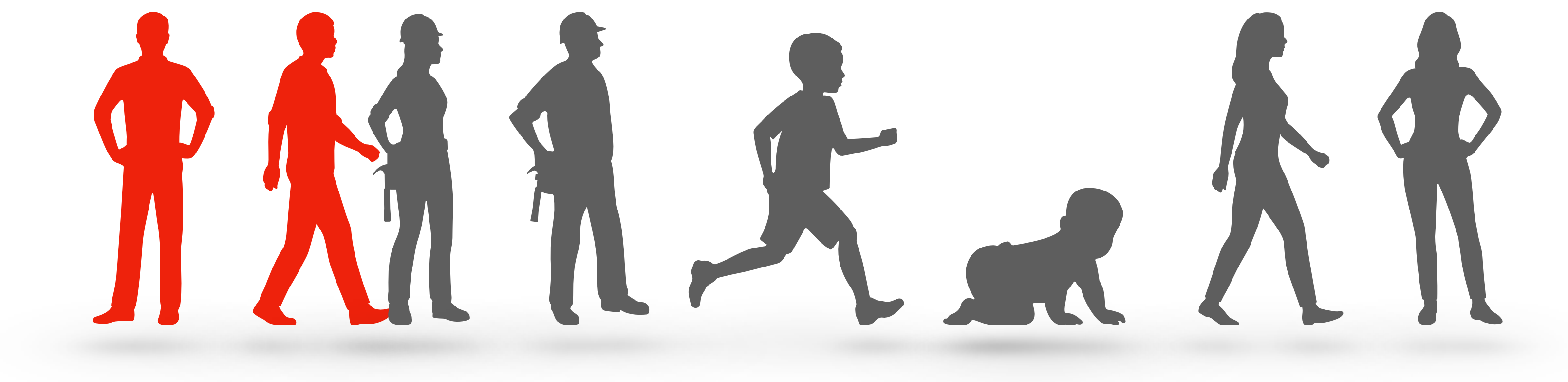
If the i^{th} way takes n_i steps then there are $n_1 + n_2 + n_3 + \cdots + n_k$ ways to do the task.

Q. Can you think of an example that uses the sum rule?

A. The number of Toronto phone numbers possible (start with 416, 647, 437).

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.



Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

$$\underline{\ 8\ } * \underline{\ 7\ } * \underline{\ 6\ } * \underline{\ 5\ } * \underline{\ 4\ } * \underline{\ 3\ } = 8!/2!$$

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

$$\underline{\ 8\ } * \underline{\ 7\ } * \underline{\ 6\ } * \underline{\ 5\ } * \underline{\ 4\ } * \underline{\ 3\ } = 8!/2!$$

Take Home

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

$\underline{\ 8\ } * \underline{\ 7\ } * \underline{\ 6\ } * \underline{\ 5\ } * \underline{\ 4\ } * \underline{\ 3\ } = 8!/2!$

Take Home

The formula to arrange *r of n* objects *without repetition* when *order matters* is:

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

$$\underline{\ 8\ } * \underline{\ 7\ } * \underline{\ 6\ } * \underline{\ 5\ } * \underline{\ 4\ } * \underline{\ 3\ } = 8!/2!$$

Take Home

The formula to arrange *r of n* objects *without repetition* when *order matters* is:

Service at the bank take 2

8 people lineup at a bank. The bank is about to close, so only 6 of the 8 people will get served.

Q. In how many different orders can 6 of the 8 people be served?

If we continue the same pattern as before we get:

$$\underline{\ 8\ } * \underline{\ 7\ } * \underline{\ 6\ } * \underline{\ 5\ } * \underline{\ 4\ } * \underline{\ 3\ } = 8!/2!$$

Take Home

The formula to arrange *r of n* objects *without repetition* when *order matters* is:

$$nPr = P(n, r) = n!/(n-r)!$$

Let's try group work!

Let’s try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Let’s try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Virtual (Zoom): We will put you in a breakout room with other students.

Let’s try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Virtual (Zoom): We will put you in a breakout room with other students.

You have three tasks:

Let’s try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Virtual (Zoom): We will put you in a breakout room with other students.

You have three tasks:

1. Find out everyone’s name and where they are from.

Let's try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Virtual (Zoom): We will put you in a breakout room with other students.

You have three tasks:

1. Find out everyone's name and where they are from.
2. Find 3 things that *everyone* in your group has in common. Must be *non-trivial* (ie, can't say you all go to UTSC, all students, all taking A67).

Let's try group work!

In Real Life: Get to know your neighbours 2-4 person groups

Virtual (Zoom): We will put you in a breakout room with other students.

You have three tasks:

1. Find out everyone's name and where they are from.
2. Find 3 things that *everyone* in your group has in common. Must be *non-trivial* (ie, can't say you all go to UTSC, all students, all taking A67).
3. Submit your group's names, where they are from and the 3 common things to our online submissions system Gradescope: https://www.gradescope.com